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NEET · PHYSICS · CHAPTER-WISE PYQS

Motion in a Plane

NEET Physics · Previous Year Questions with Answer Key

51	29	2000 – 2024	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	50 questions
MCQ (More than One Correct Answer)	1 question
Answer key	at the end

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MCQ (Single Correct Answer) 50 questions

Q1. NEET 2024 (Re-Examination)

A bob is whirled in a horizontal circle by means of a string at an initial speed of $10\,\text{rpm}$. If the tension in the string is quadrupled while keeping the radius constant, the new speed is:

Open solution

(A) 20 rpm (B) 40 rpm (C) 5 rpm (D) 10 rpm

Q2. NEET 2024 (Re-Examination)

Let ω_1, ω_2 and ω_3 be the angular speed of the second hand, minute hand and hour hand of a smoothly running analog clock, respectively. If x_1, x_2 and x_3 are their respective angular distances in 1 minute then the factor which remains constant (k) is

Open solution

(A) $\frac{\omega_1}{x_1} = \frac{\omega_2}{x_2} = \frac{\omega_3}{x_3} = k$

(B) $\omega_1 x_1 = \omega_2 x_2 = \omega_3 x_3 = k$

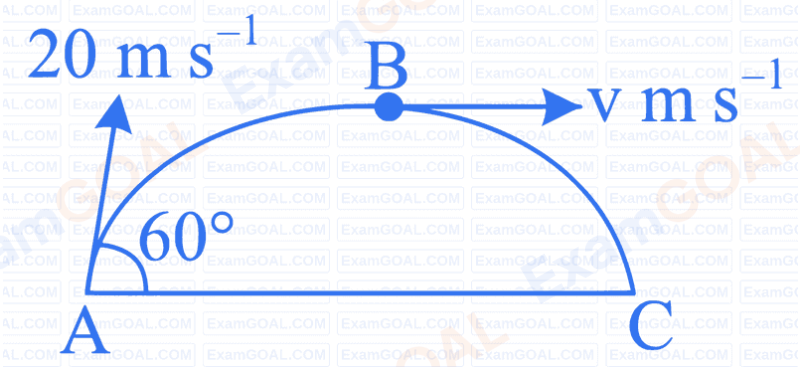
(C) $\omega_1 x_1^2 = \omega_2 x_2^2 = \omega_3 x_3^2 = k$

(D) $\omega_1^2 x_1 = \omega_2^2 x_2 = \omega_3^2 x_3 = k$

Q3. NEET 2023 Manipur

A ball is projected from point A with velocity $20\,\text{m s}^{-1}$ at an angle 60° to the horizontal direction. At the highest point B of the path (as shown in figure), the velocity vms^{-1} of the ball will be:

Open solution



(A) 20 (B) $10\sqrt{3}$ (C) Zero (D) 10

Q4. NEET 2023 Manipur

A particle is executing uniform circular motion with velocity $\vec{v}$ and acceleration $\vec{a}$. Which of the following is true?

Open solution

(A) $\vec{v}$ is a constant; $\vec{a}$ is not a constant

(B) $\vec{v}$ is not a constant; $\vec{a}$ is not a constant

(C) $\vec{v}$ is a constant; $\vec{a}$ is a constant

(D) $\vec{v}$ is not a constant; $\vec{a}$ is a constant

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Q5. NEET 2023

A bullet is fired from a gun at the speed of $280\ \text{ms}^{-1}$ in the direction 30° above the horizontal. The maximum height attained by the bullet is

$\left(g = 9.8\ \text{ms}^{-2}, \sin 30^\circ = 0.5\right)$:-

Open solution

(A) 2000 m (B) 1000 m (C) 3000 m (D) 2800 m

Q6. NEET 2023

A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity $4\ \text{m s}^{-1}$. The ball strikes the water surface after $4\ \text{s}$. The height of bridge above water surface is (Take $g = 10\ \text{m s}^{-2}$)

Open solution

(A) 60 m (B) 64 m (C) 68 m (D) 56 m

Q7. NEET 2022 Phase 2

A cricket ball is thrown by a player at a speed of 20 m/s in a direction 30° above the horizontal. The maximum height attained by the ball during its motion is

(g = 10 m/s^2)

Open solution

(A) 25 m (B) 5 m (C) 10 m (D) 20 m

Q8. NEET 2022 Phase 1

A ball is projected with a velocity, 10 ms^{-1}, at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be

Open solution

(A) Zero (B) $5\sqrt{3}\ \text{ms}^{-1}$ (C) 5 ms^{-1} (D) 10 ms^{-1}

Q9. NEET 2021

A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

Open solution

If this particle were projected with the same speed at an angle 'θ' to the horizontal, the maximum height attained by it equals 4R. The angle of projection, θ, is then given by :

(A) $\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{1/2}$ (B) $\theta = \cos^{-1}\left(\frac{gT^2}{\pi^2 R}\right)^{1/2}$

(C) $\theta = \cos^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$ (D) $\theta = \sin^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$

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Q10. NEET 2021

A car starts from rest and accelerates at 5m/s^2. At t = 4 s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at t = 6 s? (Take g = 10 m/s^2)

Open solution

(A) $20\sqrt{2}$ m/s, 10 m/s^2
(B) 20 m/s, 5 m/s^2
(C) 20 m/s, 0
(D) $20\sqrt{2}$ m/s, 0

Q11. NEET 2019

The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by :

Open solution

(A) 0°
(B) 60°
(C) 45° west
(D) 30° west

Q12. NEET 2019

When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object is shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be :

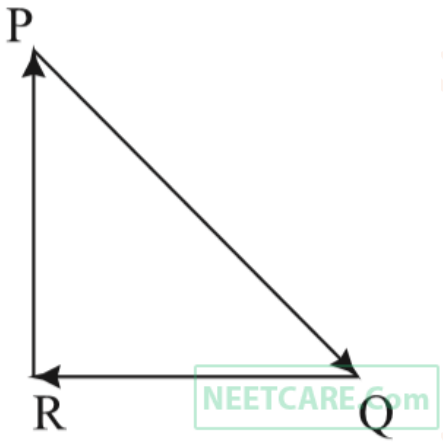
Open solution

(A) $1 : \sqrt{2}$
(B) $\sqrt{2} : 1$
(C) $1 : \sqrt{3}$
(D) $1 : 2\sqrt{3}$

Q13. NEET 2019

A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will :

Open solution



(A) remain constant
(B) increase
(C) decrease
(D) change according to the smallest force $\overrightarrow{QR}$

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Q14. NEET 2017

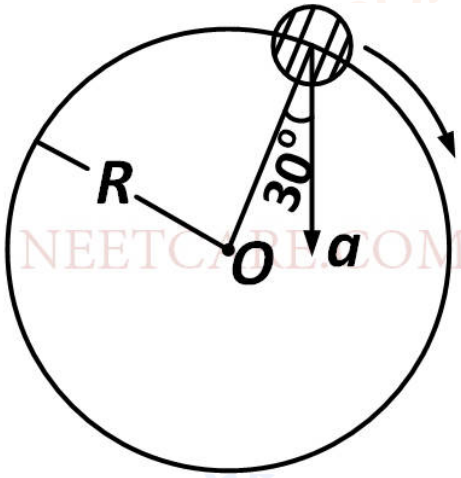
The x and y coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$ respectively, where x and y are in metres and t in seconds. The acceleration of the particle at t = 2 s is

(A) $5\ m\ s^{-2}$ (B) $-4\ m\ s^{-2}$ (C) $-8\ m\ s^{-2}$ (D) 0

Open solution

Q15. NEET 2016 Phase 2

In the given figure, $a = 15\ m\ s^{-2}$ represents the total acceleration of particle moving in the clockwise direction in a circle of radius R = 2.5 m at a given instant of time. The speed of the particle is



(A) $4.5\ m\ s^{-1}$ (B) $5.0\ m\ s^{-1}$ (C) $5.7\ m\ s^{-1}$ (D) $6.2\ m\ s^{-1}$

Open solution

Q16. NEET 2016 Phase 1

If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is

(A) 45° (B) 180° (C) 0° (D) 90°

Open solution

Q17. NEET 2016 Phase 1

A particle moves so that its position vector is given by $\vec{r} = \cos \omega t\, \hat{x} + \sin\, \omega t\, \hat{y}$, where ω is a constant.

Which of the following is true?

(A) Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin.

(B) Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin.

(C) Velocity and acceleration both are perpendicular to $\vec{r}$

(D) Velocity and acceleration both are parallel to $\vec{r}$

Open solution

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Q18. AIPMT 2015

If vectors $\vec{A} = \cos\omega t\hat{i} + \sin\omega t\hat{j}$ and $\vec{B} = \cos\frac{\omega t}{2}\hat{i} + \sin\frac{\omega t}{2}\hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other is

(A) $t = \frac{\pi}{\omega}$ (B) t = 0 (C) $t = \frac{\pi}{4\omega}$ (D) $t = \frac{\pi}{2\omega}$

Open solution

Q19. AIPMT 2015

The position vector of a particle $\vec{R}$ as a function of time is given by $\vec{R}$ = 4sin(2πt)$\hat{i}$ + 4cos(2πt)$\hat{j}$. Where R is in meters, t is in seconds and $\hat{i}$ and $\hat{j}$ denote unit vectors along x-and y-directions, respectively. Which one of the following statements is wrong for the motion of particle?

(A) Magnitude of the velocity of particle is 8 meter/second.

(B) Path of the particle is a circle of radius 4 meter.

(C) Acceleration vector is along $-\vec{R}$.

(D) Magnitude of acceleration vector is $\frac{v^2}{R}$ where v is the velocity of particle.

Open solution

Q20. AIPMT 2015 Cancelled Paper

A ship A is moving Westwards with a speed of 10 km h^{-1} and a ship B 100 km South of A, is moving Northwards with a speed of 10 km h^{-1}. The time after which the distance between them becomes shortest, is

(A) $5\sqrt{2}$ h (B) $10\sqrt{2}$ h (C) 0 h (D) 5 h

Open solution

Q21. AIPMT 2014

A projectile is fired from the surface of the earth with a velocity of 5 m s^{-1} and angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 m s^{-1} at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in m s^{-2}) is
(Given g = 9.8 m s^{-2})

(A) 3.5 (B) 5.9 (C) 16.3 (D) 110.8

Open solution

Q22. AIPMT 2014

A particle is moving such that its position coordinates (x, y) are (2 m, 3 m) at time t = 0, (6 m, 7 m) at time t = 2 s and (13 m, 14 m) at time t = 5 s.

Average velocity vector $\left(\vec{v}_{av}\right)$ from t = 0 to t = 5 s is

(A) $\frac{1}{5}\left(13\hat{i} + 14\hat{j}\right)$ (B) $\frac{7}{3}\left(\hat{i} + \hat{j}\right)$

(C) $2\left(\hat{i} + \hat{j}\right)$ (D) $\frac{11}{5}\left(\hat{i} + \hat{j}\right)$

Open solution

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Q23. NEET 2013 (Karnataka)

Vectors $\vec{A}, \vec{B}$ and $\vec{C}$ are such that $\vec{A}.\vec{B} = 0$ and $\vec{A}.\vec{C} = 0$. Then the vector parallel to $\vec{A}$ is

(A) $\vec{A} \times \vec{B}$

(B) $\vec{B} + \vec{C}$

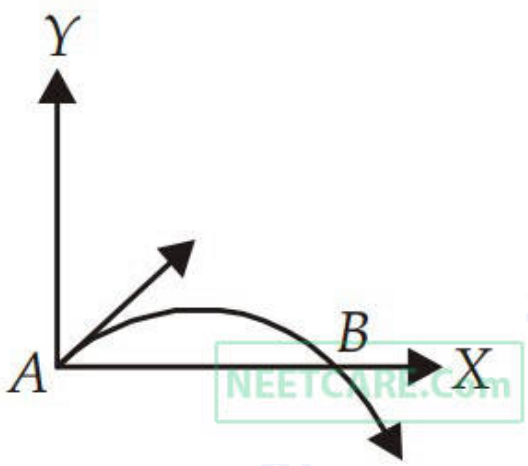
(C) $\vec{B} \times \vec{C}$

(D) $\vec{B}$ and $\vec{C}$

Open solution

Q24. NEET 2013

The velocity of a projectile at the initial point A is $\left(2\hat{i} + 3\hat{j}\right)$ m/s. Its velocity (in m/s) at point B is



(A) $2\hat{i} - 3\hat{j}$

(B) $2\hat{i} + 3\hat{j}$

(C) $-2\hat{i} - 3\hat{j}$

(D) $-2\hat{i} + 3\hat{j}$

Open solution

Q25. AIPMT 2012 Prelims

The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile is

(A) $\theta = \tan^{-1}\left(\frac{1}{4}\right)$

(B) $\theta = \tan^{-1}(4)$

(C) $\theta = \tan^{-1}(2)$

(D) $\theta = 45^{o}$

Open solution

Q26. AIPMT 2012 Prelims

A particle has initial velocity $\left(2\vec{i} + 3\vec{j}\right)$ and acceleration $\left(0.3\vec{i} + 0.2\vec{j}\right)$. The magnitude of velocity after 10 seconds will be

(A) $9\sqrt{2}$

(B) $5\sqrt{2}$

(C) 5 units

(D) 9 units

Open solution

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Q27. AIPMT 2011 Mains

A projectile is fired at an angle of 45° with the horizontal. Elevation angle of the projectile at its highest point as seen from the point of projection, is

(A) 45° (B) 60°

(C) $\tan^{-1}\left(\frac{1}{2}\right)$ (D) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Open solution

Q28. AIPMT 2011 Prelims

A particle moves in a circle of radius 5 cm with constant speed and time period 0.2π s. The acceleration of the particle is

(A) 15 m/s^2 (B) 25 m/s^2 (C) 36 m/s^2 (D) 5 m/s^2

Open solution

Q29. AIPMT 2011 Prelims

A missile is fired for maximum range with an initial velocity of 20 m/s. If g = 10 m/s^2, the range of the missile is

(A) 40 m (B) 50 m (C) 60 m (D) 20 m

Open solution

Q30. AIPMT 2011 Prelims

A body is moving with velocity 30 m/s towards east . After 10 seconds its velocity becomes 40 m/s towards north. The average acceleration of the body is

(A) 1 m/s^2 (B) 7 m/s^2 (C) $\sqrt{7}$ m/s^2 (D) 5 m/s^2

Open solution

Q31. AIPMT 2010 Mains

The speed of a projectile at its maximum height is half of its initial speed. The angle of projection is

(A) 60° (B) 15° (C) 30° (D) 45°

Open solution

Q32. AIPMT 2010 Mains

A particle moves in x-y plane according to rule x = asinωt and y = acosωt. The particle follows

(A) an elliptical path

(B) a circular path

(C) a parabolic path

(D) a straight line path inclined equally to x and y-axes

Open solution

Q33. AIPMT 2010 Prelims

A particle has initial velocity $\left(3\hat{i}+4\hat{j}\right)$ and has acceleration $\left(0.4\hat{i}+0.3\hat{j}\right)$. Its speed after 10 s is

(A) 7 units (B) $7\sqrt{2}$ units (C) 8.5 units (D) 10 units

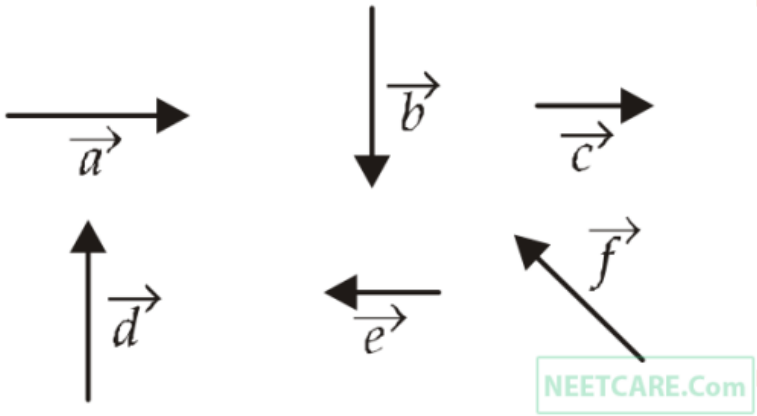
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Q34. AIPMT 2010 Prelims

Six vectors, $\vec{a}$ through $\vec{f}$ have the magnitudes and directions indicated in the figure. Which of the following statements is true ?

Open solution



(A) $\vec{b}+\vec{c}=\vec{f}$

(B) $\vec{d}+\vec{c}=\vec{f}$

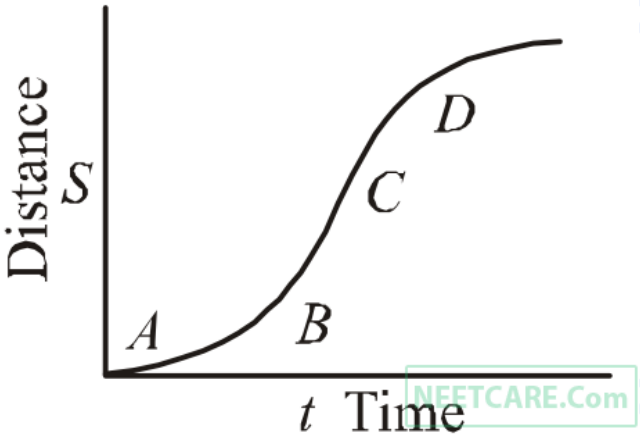
(C) $\vec{d}+\vec{e}=\vec{f}$

(D) $\vec{b}+\vec{e}=\vec{f}$

Q35. AIPMT 2008

A particle shows distance - time curve as given in this figure. The maximum instaneous velocity of the particle is around the point

Open solution



(A) D

(B) A

(C) B

(D) C

Q36. AIPMT 2007

$\vec{A}$ and $\vec{B}$ are two vectors and θ is the angle between them, if $\left|\vec{A}\times\vec{B}\right|=\sqrt{3}\left(\vec{A}.\vec{B}\right)$, the value of θ is

Open solution

(A) 45°

(B) 30°

(C) 90°

(D) 60°

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Q37. AIPMT 2007

A particle starting from the origin (0, 0) moves in a straight line in the (x, y) plane. Its coordinates at a later time are $\left(\sqrt{3}, 3\right)$. The path of the particle makes with the x-axis an angle of

(A) 45° (B) 60° (C) 0° (D) 30°.

Open solution

Q38. AIPMT 2006

For angles of projection of a projectile at angle $(45^o - \theta)$ and $(45^o + \theta)$, the horizontal range described by the projectile are in the ratio of

(A) 2 : 1 (B) 1 : 1 (C) 2 : 3 (D) 1 : 2.

Open solution

Q39. AIPMT 2006

The vectors $\vec{A}$ and $\vec{B}$ are such that $\left|\vec{A} + \vec{B}\right| = \left|\vec{A} - \vec{B}\right|$. The angle between the two vectors is

(A) 45° (B) 90° (C) 60° (D) 75°

Open solution

Q40. AIPMT 2005

Two boys are standing at the ends A and B of a ground where AB = a. The boy at B starts running in a direction perpendicular to AB with velocity v_1. The boy at A starts running simultaneously with velocity v and catches the other in a time t, where t is

(A) $\frac{a}{\sqrt{v^2 + {v_1}^2}}$ (B) $\frac{a}{v + v_1}$

(C) $\frac{a}{v - v_1}$ (D) $\sqrt{\frac{a^2}{v^2 - {v_1}^2}}$

Open solution

Q41. AIPMT 2005

If the angle between the vectors $\vec{A}$ and $\vec{B}$ is θ, the value of the product $\left(\vec{B} \times \vec{A}\right).\vec{A}$ is equal to

(A) $BA^2\sin\theta$ (B) $BA^2\cos\theta$

(C) $BA^2\sin\theta\cos\theta$ (D) zero.

Open solution

Q42. AIPMT 2005

If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$, then the value of α is

(A) 1/2 (B) $-$ 1/2 (C) 1 (D) $-$ 1.

Open solution

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Q43. AIPMT 2004

If $\left|\vec{A}\times\vec{B}\right| = \sqrt{3}\vec{A}.\vec{B}$ then the value of $\left|\vec{A}+\vec{B}\right|$ is

(A) $(A^2 + B^2 + AB)^{1/2}$

(B) $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$

(C) A + B

(D) $(A^2 + B^2 + \sqrt{3}AB)^{1/2}$.

Open solution

Q44. AIPMT 2003

The vector sum of two forces is perpendicular to their vector differences. In that case, the forces

(A) are equal to each other

(B) are equal to each other in magnitude

(C) are not equal to each other in magnitude

(D) cannot be predicted.

Open solution

Q45. AIPMT 2002

A particle A is dropped from a height and another particle B is projected in horizontal direction with speed of 5 m/sec from the same height then correct statement is

(A) particle A will reach at ground first with respect to particle B

(B) particle B will reach at ground first with respect to particle A

(C) both particles will reach at ground simultaneously

(D) both particles will reach at ground with same speed.

Open solution

Q46. AIPMT 2001

If $\left|\vec{A}+\vec{B}\right| = \left|\vec{A}\right| + \left|\vec{B}\right|$ then angle between A and B will be

(A) 90°

(B) 120°

(C) 0°

(D) 60°.

Open solution

Q47. AIPMT 2001

Two particles having mass M and m are moving in a circular path having radius R and r. If their time period are same then the ratio of angular velocity will be

(A) $\frac{r}{R}$

(B) $\frac{R}{r}$

(C) 1

(D) $\sqrt{\frac{R}{r}}$

Open solution

Q48. AIPMT 2000

The width of river is 1 km. The velocity of boat is 5 km/hr. The boat covered the width of river along the shortest path in 15 min. Then the velocity of river stream is

(A) 3 km/hr

(B) 4 km/hr

(C) $\sqrt{29}$ km/hr

(D) $\sqrt{41}$ km/hr.

Open solution

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Q49. AIPMT 2000

Two projectiles of same mass and with same velocity are thrown at an angle 60° and 30° with the horizontal, then which will remain same

(A) time of flight

(B) range of projectile

(C) maximum height acquired

(D) all of them.

Open solution

Q50. AIPMT 2000

A man is slipping on a frictionless inclined plane and a bag falls down from the same height. Then the velocity of both is related as

(A) $v_B > v_m$

(B) $v_B < v_m$

(C) $v_B = v_m$

(D) v_B and v_m can't be related.

Open solution

MCQ (More than One Correct Answer) 1 questions

Q51. AIPMT 2005

A stone tied to the end of a string of 1 m long is whirled in a horizontal circle with a constant speed. If the stone makes 22 revolutions in 44 seconds, what is the magnitude and direction of acceleration of the stone ?

(A) π^2 m s^{-2} and direction along the radius towards the centre

(B) π^2 m s^{-2} and direction along the radius away from the centre

(C) π^2 m s^{-2} and direction along the tangent to the circle

(D) $\pi^2/4$ ms^{-2} and direction along the radius towards the centre.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **A**	3. **D**	4. **B**	5. **B**
6. **B**	7. **B**	8. **B**	9. **A**	10. **A**
11. **D**	12. **C**	13. **A**	14. **B**	15. **C**
16. **D**	17. **A**	18. **A**	19. **A**	20. **D**
21. **A**	22. **D**	23. **C**	24. **A**	25. **B**
26. **B**	27. **C**	28. **D**	29. **A**	30. **D**
31. **A**	32. **B**	33. **B**	34. **C**	35. **D**
36. **D**	37. **B**	38. **B**	39. **B**	40. **D**
41. **D**	42. **B**	43. **A**	44. **B**	45. **C**
46. **C**	47. **C**	48. **A**	49. **B**	50. **C**

MCQ (More than One Correct Answer)

51. **A, B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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